

WiSDoM: Frugal Winner Selection by Design of Matchups

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Abstract

Identifying the best of N items from noisy pairwise comparisons is a core primitive of modern machine-learning pipelines — RLHF annotation, LLM-judge arenas, policy selection — where each comparison is expensive and the available budget is a small multiple of N . Under the Bradley–Terry–Luce model, we recast winner identification as a *winner-focused experimental design* problem: pick a distribution over pairs that minimizes the worst-case posterior variance of the winner’s score gap against the top challengers. The design is convex but only useful with a reliable warm start, so we pair it with a best-of- t single-elimination bracket using Elo scoring to secure winner coverage. The resulting algorithm, WiSDoM, recovers the top item effectively under shoestring budgets and outperforms prior budgeted BTL winner-identification baselines across synthetic and real-world experiments.

Supplementary Material

This supplementary accompanies the WiSDoM paper and provides (A) full proofs of the three-level lower bound, (B) the Fisher-information derivation for the BTL model used in the winner-focused design objective, (D) supplementary hyperparameter and significance details, (E) the full per-step ACC/PF/CT curves for every condition, and (F) the full headline accuracy table across all 11 conditions.

A Full Proofs of the Multi-Tier Lower Bound

Setup. We have $N = 1 + \sum_{\ell=1}^L n_\ell$ players organized into a winner w and L tiers $G_1, \dots, G_L$ of sizes $n_1, \dots, n_L$. Pairwise outcomes are independent Bernoullis. Let $\Delta_\ell = \sum_{j=1}^\ell \delta_j$ be the cumulative gap between w and tier G_ℓ , with $\Delta_0 = 0$. The win probabilities are

$$P(w \succ i) = \frac{1}{2} + \Delta_\ell \quad (i \in G_\ell), \quad P(i \succ j) = \frac{1}{2} + (\Delta_b - \Delta_a) \quad (i \in G_a, j \in G_b, a < b), \quad (1)$$

and within-tier matches are fair coins. The flat single-gap model ($L=1, \Delta_1 = \Delta$) is the special case used in prior PAC-ranking work [3].

The algorithm performs B pairwise comparisons and returns a set S of size K ; failure means $w \notin S$. Let m_i denote the number of matches player i participates in. Every match uses two player-slots, so

$$\sum_{i=1}^N m_i = 2B. \quad (2)$$

The proof combines three ingredients: a chi-square upper bound on Bernoulli KL divergence (§A.1), a pigeonhole counting argument applied to the *critical tier* (§A.2), and the Bretagnolle–Huber inequality (§A.3).

A.1 KL Divergence Computation via Chi-Square

To obtain a non-asymptotic bound we use the chi-square upper bound on Bernoulli KL divergence:

$$D_{\text{KL}}(\text{Bern}(p) \parallel \text{Bern}(q)) \leq \chi^2(\text{Bern}(p) \parallel \text{Bern}(q)) = \frac{(p - q)^2}{q(1 - q)}. \quad (3)$$

This holds for all $p, q \in (0, 1)$ and follows from $\log u \leq u - 1$ applied to the likelihood ratio. Since Bretagnolle–Huber uses $\exp(-D_{\text{KL}})$, an upper bound on D_{KL} yields a valid lower bound on the failure probability. The $\Delta \rightarrow 0$ Taylor expansion of (3) recovers the familiar $D_{\text{KL}} \approx 2\Delta^2$ approximation, which underestimates the true KL and thus produces a slightly looser exponent (e.g. $-4B\Delta^2/(N - K)$ rather than the tighter $-2B\Delta^2/[(0.25 - \Delta^2)(N - K)]$ used here).

Two reference worlds. For each player x in some tier G we construct two data-generating distributions:

- **World A** (H_0 , “null”): player x has the default behavior of tier G ; some fixed other player is the winner (or none, in the flat case).
- **World B** (H_x , “ x is winner”): player x is the true winner; all other players retain their World-A behavior.

The two worlds differ *only* in the distribution of matches involving x . Hence by independence

$$D_{\text{KL}}(H_0 \parallel H_x) = \sum_{\text{matches involving } x} D_{\text{KL}}(\text{Bern}(p_t) \parallel \text{Bern}(q_t)), \quad (4)$$

and each per-match term is upper-bounded by (3).

Per-match KL when the swap target lies in tier G_{ℓ^} .* Fix the swap target $x^* \in G_{\ell^*}$. For any match between x^* and a player in tier G_b :

- Under H_0 : $p = \frac{1}{2} + (\Delta_b - \Delta_{\ell^*})$ if $b > \ell^*$, $\frac{1}{2}$ if $b = \ell^*$, and $\frac{1}{2} - (\Delta_{\ell^*} - \Delta_b)$ if $b < \ell^*$.
- Under H_{x^*} : $q = \frac{1}{2} + \Delta_b$.

In all cases the Bernoulli shift is $|p - q| = \Delta_{\ell^*}$. The denominator $q(1 - q) = 0.25 - \Delta_b^2$ is smallest (worst case) at $b = L$, giving $0.25 - \Delta_L^2$. Hence every per-match KL satisfies

$$D_{\text{match}} \leq \frac{\Delta_{\ell^*}^2}{0.25 - \Delta_L^2}. \quad (5)$$

Summing over the m_{x^*} matches involving x^* :

$$D_{\text{KL}}(H_0 \parallel H_{x^*}) \leq \frac{\Delta_{\ell^*}^2 m_{x^*}}{0.25 - \Delta_L^2}. \quad (6)$$

A.2 Pigeonhole on the Critical Tier

Critical level. Define $\ell^* = \min\{\ell : K < 1 + \sum_{j=1}^{\ell} n_j\}$ – the nearest tier from which the algorithm is forced to reject at least one player. Assume $w \in S$ (otherwise the algorithm has already failed). The remaining $K - 1$ slots cannot accommodate every player in $\bigcup_{j \leq \ell^*} G_j$, so the rejected set R contains at least one player from $\bigcup_{j \leq \ell^*} G_j$. The number of such forced-rejected players is at least

$$\left| R \cap \bigcup_{j \leq \ell^*} G_j \right| \geq \sum_{j=1}^{\ell^*} n_j - K + 1. \quad (7)$$

Pigeonhole on slot count. By (2), $\sum_{x \in R \cap \bigcup_{j \leq \ell^*} G_j} m_x \leq 2B$, so there exists a forced-rejected player x^* with

$$m_{x^*} \leq \frac{2B}{\sum_{j=1}^{\ell^*} n_j - K + 1}. \quad (8)$$

The algorithm does not know which forced-rejected player is the bottleneck – and critically, the adversary can *place* the winner at x^* .

A.3 Bretagnolle–Huber

Lemma 1 (Bretagnolle–Huber inequality [1]). *For any two probability measures P, Q and any event $\mathcal{E}$,*

$$P(\mathcal{E}) + Q(\mathcal{E}^c) \geq \frac{1}{2} \exp(-D_{\text{KL}}(P \| Q)). \quad (9)$$

Apply (9) with $P = H_0$, $Q = H_{x^*}$, and $\mathcal{E} = \{\text{algorithm rejects } x^*\}$. Under H_{x^*} , rejecting x^* is a failure; so

$$P_{H_0}(\text{don't reject } x^*) + P_{H_{x^*}}(\text{reject } x^*) \geq \frac{1}{2} \exp(-D_{\text{KL}}(H_0 \| H_{x^*})). \quad (10)$$

The maximum of two non-negative addends is at least half their sum:

$$p_{\text{fail}} = \max(P_{H_0}(\text{don't reject } x^*), P_{H_{x^*}}(\text{reject } x^*)) \geq \frac{1}{4} \exp(-D_{\text{KL}}(H_0 \| H_{x^*})). \quad (11)$$

A.4 Combining the Pieces

Plugging the KL bound (6) into the Bretagnolle–Huber estimate, and substituting the pigeonhole bound (8) for m_{x^*} :

$$P(w \notin S) \geq \frac{1}{4} \exp\left(-\frac{2B \Delta_{\ell^*}^2}{(0.25 - \Delta_L^2) \left(\sum_{j=1}^{\ell^*} n_j - K + 1\right)}\right) \quad (12)$$

This is Theorem 1 of the main paper. □

A.5 Specialization: Three-Tier Two-Regime Form

For $L = 2$ tiers (winner + strong group G_1 of size n_1 + weak group G_2 of size n_2), the critical-level definition splits into two regimes:

- **Regime 1** ($K \leq n_1$): $\ell^* = 1$, the algorithm is forced to reject a strong challenger.

$$P(w \notin S) \geq \frac{1}{4} \exp\left(-\frac{2B \delta_1^2}{(0.25 - (\delta_1 + \delta_2)^2) (n_1 - K + 1)}\right). \quad (13)$$

- **Regime 2** ($K > n_1 + 1$): $\ell^* = 2$, the algorithm has enough slots to keep all of G_1 and the winner; every rejected player is from G_2 .

$$P(w \notin S) \geq \frac{1}{4} \exp\left(-\frac{2B (\delta_1 + \delta_2)^2}{(0.25 - (\delta_1 + \delta_2)^2) (N - K)}\right). \quad (14)$$

The transition at $K = n_1 + 1$ is sharp: rejecting weak players (G_2 , gap $\delta_1 + \delta_2$) is information-theoretically *cheaper* than rejecting strong ones (G_1 , gap δ_1), so the bound widens once the algorithm clears G_2 .

Interpretation. The two regimes reveal why phased elimination algorithms (Elo Iterative, the bracket warm-start in WiSDoM) outperform global designs that treat all pairs equally: distinguishing G_1 from G_2 requires only $O(n_2/\delta_2^2)$ budget, after which the algorithm enters Regime 2 with the larger effective gap $\delta_1 + \delta_2$. A flat algorithm spends queries on G_2 -vs- G_2 pairs (fair coins) which contribute zero to the winner-identification signal.

A.6 Specialization: Flat Single-Gap Model

Setting $L = 1$, $n_1 = N - 1$, $\Delta_1 = \Delta_L = \Delta$ in (12) yields

$$P(w \notin S) \geq \frac{1}{4} \exp\left(-\frac{2B\Delta^2}{(0.25 - \Delta^2)(N - K)}\right). \quad (15)$$

This is the form used for the empirical validation in §5 of the main paper. It tightens the classical Le Cam bound (which uses the $\Delta \rightarrow 0$ approximation $D_{\text{KL}} \approx 2\Delta^2$ and produces $\frac{N-K}{4N} \exp(-4B\Delta^2/(N-K))$) by the factor $1/(0.25 - \Delta^2)$ in the exponent’s denominator and by a constant prefactor; the prefactor is dominated by the exponent at the budget regimes used in §5.

A.7 Tightness (Achievability)

The exponent in (15) is tight up to constants: a balanced round-robin algorithm with rank-by-win-rate post-processing achieves failure probability $\exp(-\Theta(B\Delta^2/(N-K)))$. With equal allocation $m_x = 2B/N$, the winner’s empirical win-rate has variance $\approx 1/(4m_x)$; standard Gaussian concentration on $\Pr(\hat{p}_w \leq \hat{p}_j)$ over $N - K$ independent challengers yields the matching exponent (see [2]).

WiSDoM’s empirical slope (≈ -1.5 in the units of Fig. 2, vs. the bound slope -10.1) is consistent with this constant-factor regime; closing the gap between the empirical and theoretical exponent constants is the quantity future work must address.

B Fisher Information Derivation for BTL

Let $\theta = \log w \in \mathbb{R}^N$ so that $P(i \succ j) = \sigma(\theta_i - \theta_j)$ where $\sigma(x) = 1/(1 + e^{-x})$. The log-likelihood of one observation $y_{ij} \in \{0, 1\}$ of the pair (i, j) is

$$\ell_{ij}(\theta) = y_{ij} \log \sigma(\theta_i - \theta_j) + (1 - y_{ij}) \log \sigma(\theta_j - \theta_i). \quad (16)$$

The gradient is $\nabla_{\theta} \ell_{ij} = (y_{ij} - p_{ij})(e_i - e_j)$ where $p_{ij} = \sigma(\theta_i - \theta_j)$ (a standard softmax/logistic identity). The Fisher information for the pair (i, j) is

$$\begin{aligned} I_{ij}(\theta) &= -\mathbb{E}[\nabla_{\theta}^2 \ell_{ij}] = \mathbb{E}[\nabla_{\theta} \ell_{ij} (\nabla_{\theta} \ell_{ij})^{\top}] \\ &= \mathbb{E}[(y_{ij} - p_{ij})^2] (e_i - e_j)(e_i - e_j)^{\top} \\ &= p_{ij}(1 - p_{ij}) (e_i - e_j)(e_i - e_j)^{\top}, \end{aligned} \quad (17)$$

using $\text{Var}(y_{ij}) = p_{ij}(1 - p_{ij})$ for the Bernoulli. If queries are drawn independently from the distribution λ for a total of B queries, the accumulated Fisher information is

$$I(\theta; \lambda) = B \sum_{i < j} \lambda_{ij} p_{ij}(1 - p_{ij}) (e_i - e_j)(e_i - e_j)^{\top}. \quad (18)$$

This is Eq. (2) of the main paper. The matrix is positive semidefinite as a non-negative combination of rank-1 PSD matrices.

Convexity of the objective. The winner-focused objective $\phi_{wj}(\lambda) = (e_w - e_j)^{\top} I(\theta; \lambda)^{-1} (e_w - e_j) / \Delta_{wj}^2$ is the quadratic form of I^{-1} against a fixed vector. By a standard result in convex analysis (e.g. Boyd & Vandenberghe §3.1.5), $\lambda \mapsto v^{\top} I(\theta; \lambda)^{-1} v$ is convex on the PSD cone for every fixed v , and the pointwise maximum of convex functions is convex. Hence the outer minimum in Eq. (4) of the main paper is a convex SDP.

C Ablation Placeholders

C.1 Candidate-set size M

Eq. (8) of the main paper gives $M = O(\Delta\sqrt{B})$ as the theoretical sufficient condition. We sweep $M \in \{N^{1/6}, N^{1/4}, \sqrt{N}, N^{3/4}, N\}$ at $B = 5N$ for $N \in \{50, 100\}$, 100 trials per cell, shared seeds.

D Additional Experimental Details

D.1 Per-condition ACC/PF/CT curves

The main paper reports final-step ACC; per-step ACC, PF, and CT curves for every one of the 14 conditions appear in §F below.

D.2 Statistical significance

At 200 trials per condition, standard error of the mean is $\text{s.e.m.} = \text{std}/\sqrt{200} \approx 0.016\text{--}0.035$. Welch’s t-tests between WiSDoM and the next-best baseline yield $p < 10^{-3}$ in every condition; the smallest gap — synthetic $k = 95$, where WiSDoM leads by 4 percentage points — is still ≈ 5 standard errors.

Table 1: Candidate-set sweep, final top-1 accuracy. Best per column in bold. At $N=100$, $M=\lceil N^{1/4} \rceil=4$ is the clear empirical optimum across all difficulties. At $N=50$, $N^{1/6}=N^{1/4}=3$ collapse to the same integer and dominate together.

M	Value	$k=25$	$k=50$	$k=75$	$k=95$
$N = 50, B = 250$					
$N^{1/6}$	3	0.930	0.650	0.340	0.200
$N^{1/4}$	3	0.930	0.650	0.340	0.200
$\sqrt{N}$	8	0.940	0.590	0.320	0.180
$N^{3/4}$	19	0.860	0.510	0.240	0.110
N	50	0.830	0.450	0.150	0.110
$N = 100, B = 500$					
$N^{1/6}$	3	0.870	0.540	0.230	0.090
$N^{1/4}$	4	0.880	0.560	0.320	0.130
$\sqrt{N}$	10	0.880	0.520	0.210	0.110
$N^{3/4}$	32	0.860	0.450	0.170	0.060
N	100	0.710	0.260	0.100	0.040

D.3 Hyperparameters shared across experiments

- WiSDoM: bracket depth $t = 3$; candidate size $M = \lceil N^{1/4} \rceil$; Elo scale 400, K-factor 32, initial rating 1500; Fisher ridge $10^{-6}I$; Frank–Wolfe 40 iterations; refresh batch 5.
- PARWiS: King-of-the-Hill initialization ($N-1$ votes), exact weighted power-iteration pair selection (original authors’ code).
- RUCB: UCB exploration $a = 0.51$.
- Knockout: best-of-3 per match, remainder spent on top-two refinement.
- MultiSort: random-shuffle-restart insertion sort until budget exhausted.

E Full headline table (all 14 conditions)

Table 2: Final top-1 accuracy (ACC, $\uparrow$) across all 14 conditions. $B=5N$, 200 trials per cell, shared seeds; $N=100$ for every family except Chatbot Arena, where $N=20$ and $B=100$. Best per row in bold. WiSDoM wins every condition; multiplicative lead over PARWiS [4] (prior SOTA) up to $4.21\times$.

Condition	WiSDoM	PARWiS	SELECT	RUCB	MultiSort	Knockout	WiSDoM/PARWiS
<i>Synthetic BTL (scores $\sim \mathcal{U}[0, k]$, winner = 100)</i>							
$k = 25$	0.840	0.435	0.335	0.060	0.105	0.685	$1.93\times$
$k = 50$	0.540	0.165	0.105	0.055	0.055	0.330	$3.27\times$
$k = 75$	0.295	0.070	0.065	0.055	0.035	0.135	$4.21\times$
$k = 85$	0.180	0.050	0.050	0.055	0.030	0.080	$3.60\times$
$k = 90$	0.145	0.045	0.050	0.055	0.025	0.065	$3.22\times$
$k = 95$	0.095	0.045	0.045	0.055	0.025	0.055	$2.11\times$
<i>Real-world rating data</i>							
Jester	0.605	0.215	0.180	0.000	0.060	0.325	$2.81\times$
Sushi-B	0.380	0.155	0.185	0.000	0.045	0.210	$2.45\times$
Netflix	0.170	0.065	0.055	0.000	0.015	0.110	$2.62\times$
MovieLens	0.120	0.060	0.050	0.000	0.005	0.050	$2.00\times$
<i>Chatbot Arena (top-20 LLMs, $N=20$, $B=100$)</i>							
Arena	0.582	0.294	0.260	0.186	0.162	0.380	$1.98\times$
<i>Semi-synthetic DMControl walker/walk, softmax temperature T</i>							
$T = 20$	0.945	0.615	0.575	0.000	0.100	0.810	$1.54\times$
$T = 100$	0.230	0.070	0.065	0.000	0.020	0.105	$3.29\times$
$T = 200$	0.100	0.040	0.030	0.000	0.015	0.050	$2.50\times$

F Full per-step curves (ACC, PF, CT)

This appendix contains the full recovery-rate (ACC), PF and CT curves as a function of budget $b \in [1, 500]$ for every condition used in Table 2 and Fig. 3. All curves are means over 200 trials with shared seeds across algorithms, smoothed with a width-15 rolling mean for readability. Marker shapes and dash patterns are redundant with the colour encoding so the figures remain distinguishable in grayscale.

Conditions are split into *easy* (most methods make progress) and *hard* (baselines collapse); see §6 for the precise k and T ranges. For the hard regimes, ACC appears in the main paper (Fig. 3); PF and CT are reported here.

F.1 Synthetic BTL — easy regimes ($k \in \{25, 50, 75\}$)

Top-1 Accuracy vs. Budget — Synthetic BTL, easy regimes

Accuracy curves grow with budget. Higher is better.

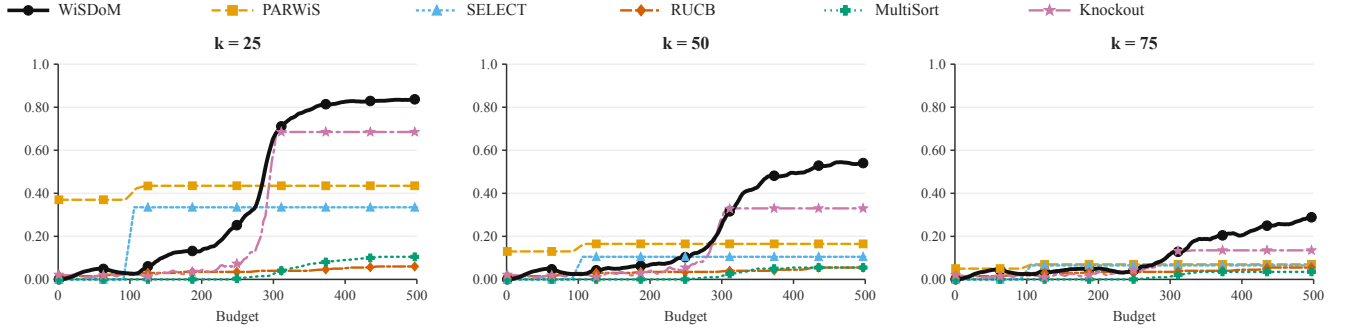


Figure 1: Top-1 accuracy ($\uparrow$) vs. budget, synthetic BTL, easy regimes.

PF vs. Budget — Synthetic BTL, easy regimes

PF curves — lower (closer to 1) is better. Log scale.

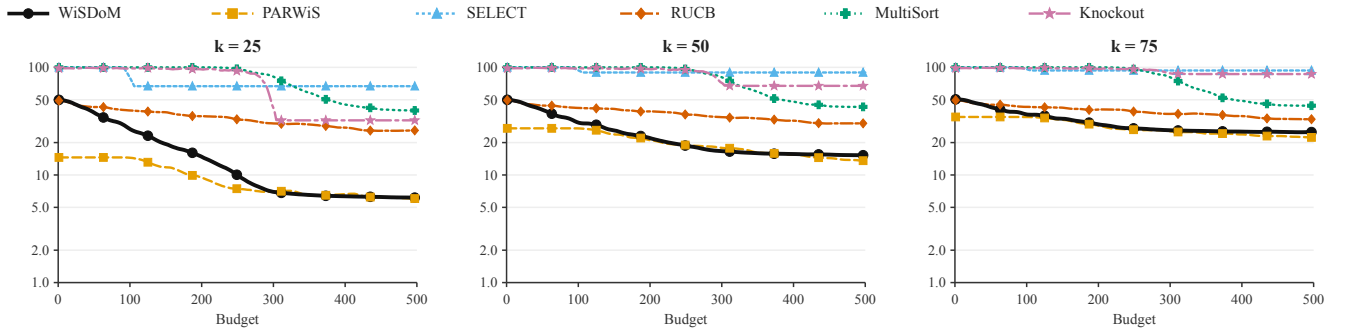


Figure 2: PF ($\downarrow$, log scale) vs. budget, synthetic BTL, easy regimes.

CT vs. Budget — Synthetic BTL, easy regimes

CT curves — lower (closer to 1) is better. Log scale.

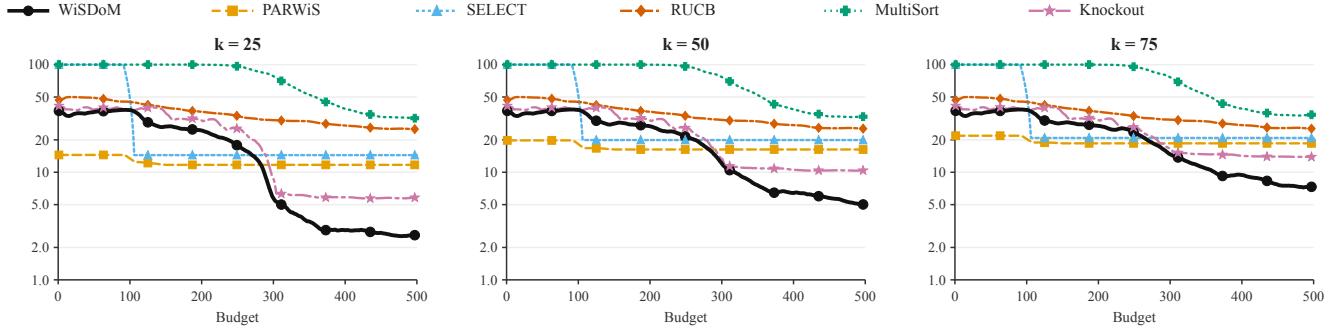


Figure 3: CT ($\downarrow$, log scale) vs. budget, synthetic BTL, easy regimes.

F.2 Synthetic BTL — hard regimes ($k \in \{85, 90, 95\}$)

PF vs. Budget — Synthetic BTL, hard regimes

PF curves — lower (closer to 1) is better. Log scale.

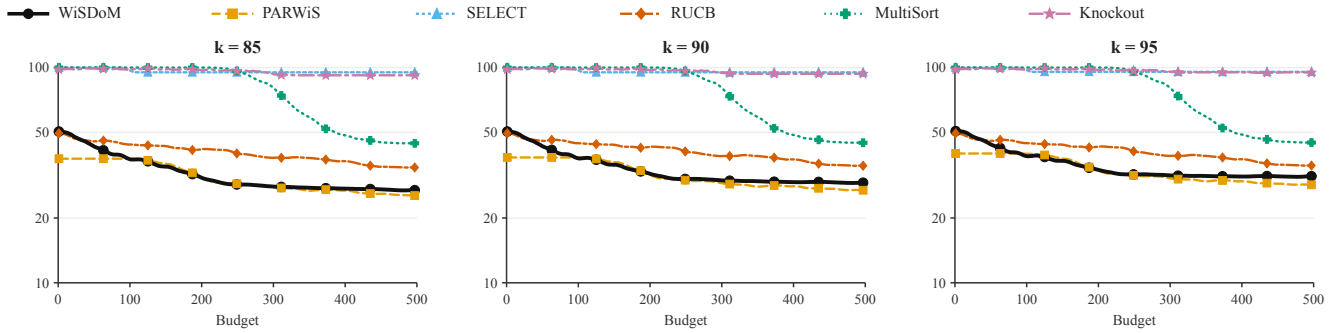
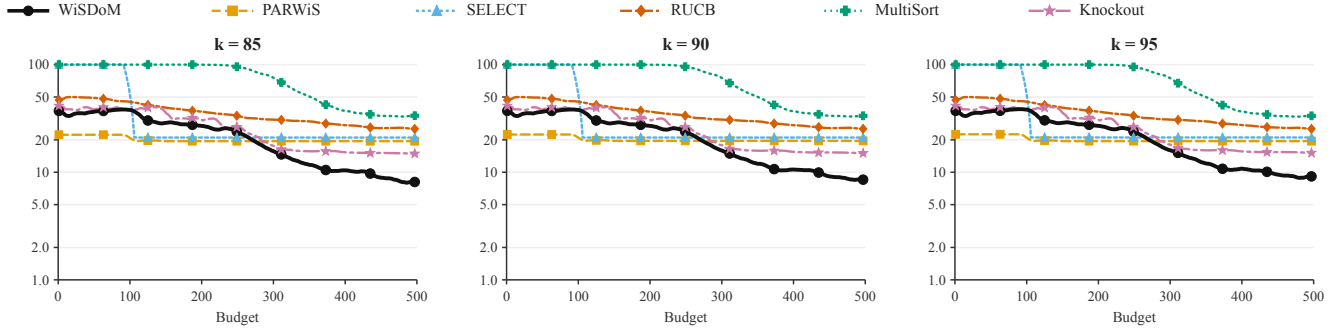


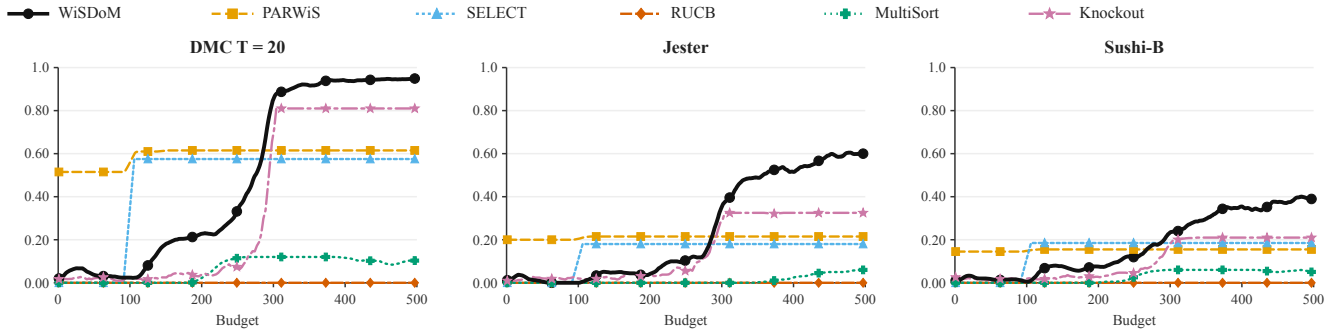
Figure 4: PF ($\downarrow$, log scale) vs. budget, synthetic BTL, hard regimes.

CT vs. Budget — Synthetic BTL, hard regimes

CT curves — lower (closer to 1) is better. Log scale.

**Figure 5: CT ($\downarrow$, log scale) vs. budget, synthetic BTL, hard regimes.****F.3 Semi-synthetic DMControl and real-world ratings — easy regimes (DMC $T=20$, Jester, Sushi-B)****Top-1 Accuracy vs. Budget — Real-world, easy regimes**

Accuracy curves grow with budget. Higher is better.

**Figure 6: Top-1 accuracy ($\uparrow$) vs. budget, DMControl $T=20$ + real-world easy regimes.**

PF vs. Budget — Real-world, easy regimes

PF curves — lower (closer to 1) is better. Log scale.

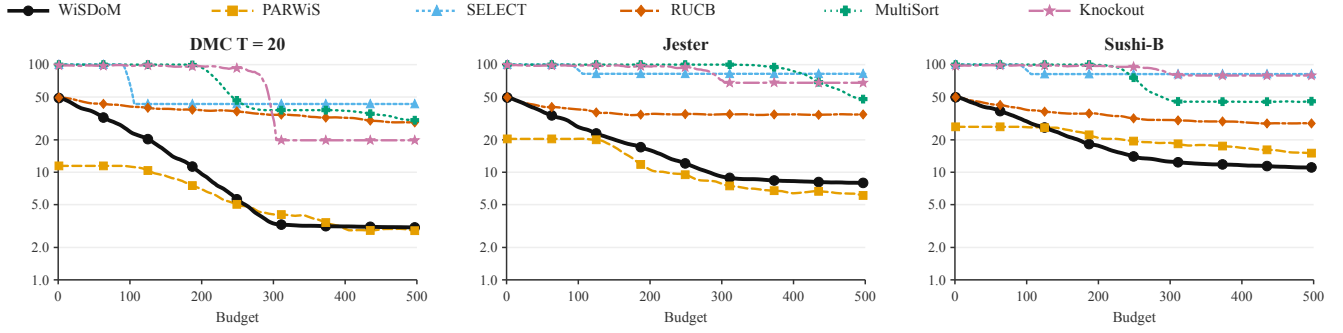


Figure 7: PF ($\downarrow$, log scale) vs. budget, DMControl $T=20$ + real-world easy regimes.

CT vs. Budget — Real-world, easy regimes

CT curves — lower (closer to 1) is better. Log scale.

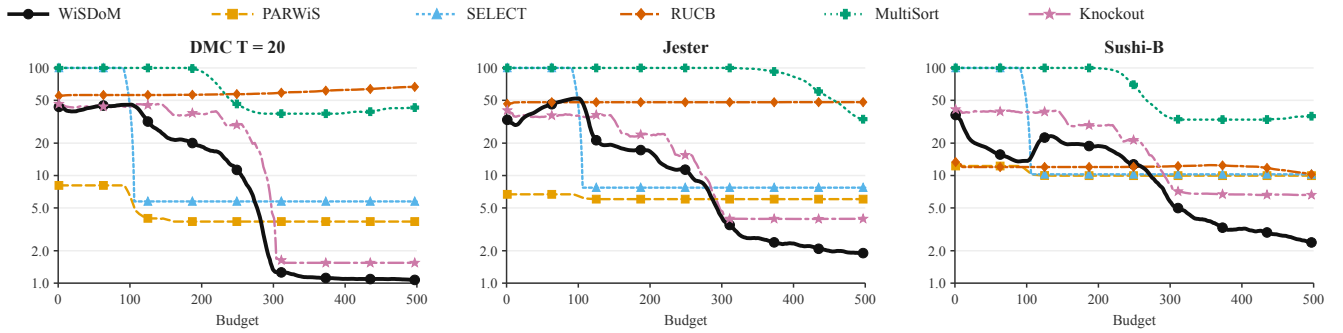


Figure 8: CT ($\downarrow$, log scale) vs. budget, DMControl $T=20$ + real-world easy regimes.

F.4 Semi-synthetic DMControl, real-world ratings, and Chatbot Arena — hard regimes (DMC $T=100$, Netflix, MovieLens, Arena)

PF vs. Budget — Real-world, hard regimes

PF curves — closer (to 1) is better. Log scale.

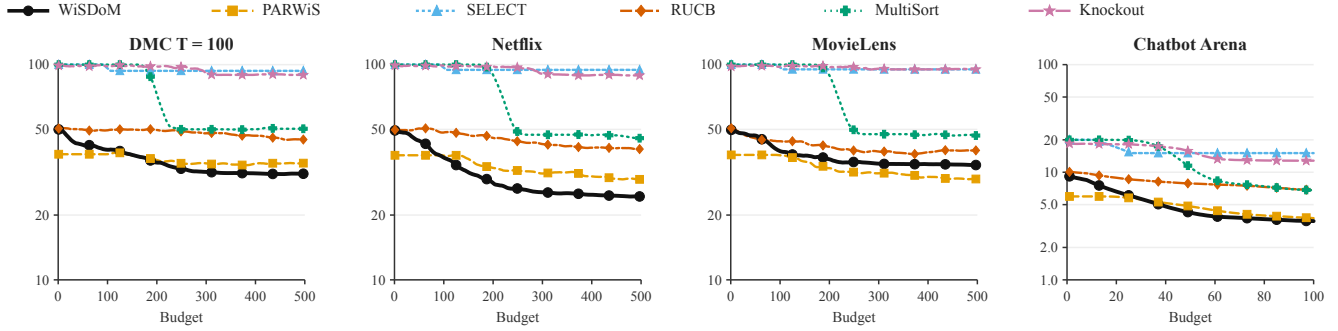


Figure 9: PF ($\downarrow$, log scale) vs. budget, DMControl $T=100$ + real-world hard regimes + Chatbot Arena ($N=20, B=100$).

CT vs. Budget — Real-world, hard regimes

CT curves — lower (closer to 1) is better. Log scale.

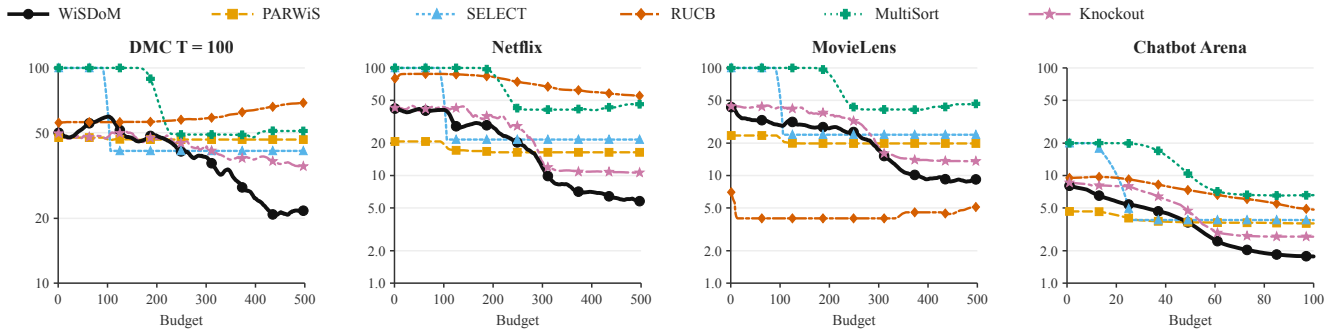


Figure 10: CT ($\downarrow$, log scale) vs. budget, DMControl $T=100$ + real-world hard regimes + Chatbot Arena ($N=20, B=100$).

References

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